Next-Day Air Quality Index Forecasting and Health-Alert Classification for 26 Indian Cities Using Classical Machine-Learning Models

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Abstract. Air pollution is a major health risk in India, yet the Air Quality Index (AQI) is normally published only for the day that has already passed. This study asks whether classical machine-learning models (four main ones per task, more in the appendix) can forecast tomorrow's AQI and warn of "Poor"-or-worse days (AQI > 200) one day in advance. We use public daily data for 26 Indian cities (2015–2020) compiled from the Central Pollution Control Board (CPCB). Models were trained on 2015–2018 (13,041 samples), tuned with five-fold forward-chaining cross-validation and tested on January 2019 to February 2020 (8,374 samples) against a persistence baseline (tomorrow equals today). For regression, a support-vector regressor (SVR) gave the lowest test RMSE (34.96; $R^2 = 0.891$), 17.6% below persistence (42.43), while plain linear regression came within 5.5% of it. For the health alert, logistic regression and an RBF support-vector machine tied for the best F1 score (0.866; ROC-AUC 0.981 for logistic regression) against 0.833 for persistence. During the 2020 COVID-19 lockdown, however, the tuned SVR (RMSE 25.18) was worse than persistence (21.11), and it also lost to persistence in 5 of 25 cities, four of them with no training data. Simple models may therefore be adequate for city-level early warning on normal days, provided they are monitored and re-trained when conditions change.

Keywords: air quality index; India; CPCB; regression; classification; support vector machine; logistic regression; time-series cross-validation; SDG 3; SDG 11

1 Introduction

Ambient air pollution is a leading cause of death and illness in India. A state-level Global Burden of Disease analysis attributed about 1.67 million deaths in 2019 to air pollution [1]. The Central Pollution Control Board (CPCB) reports a daily AQI that maps pollutant concentrations to six categories, from Good (0–50) to Severe (401–500) [2]. Because the index describes the day that has already passed, children, the elderly and people with asthma get little advance warning.

Forecasting tomorrow's AQI is therefore a socially useful machine-learning (ML) problem, linked to SDG 3 (Good Health and Well-being) and SDG 11 (Sustainable Cities and Communities). Many recent studies use complex ensembles or deep networks. We instead ask whether four classical models taught in an undergraduate course are already enough when they are evaluated against a simple baseline.

Our contributions are: (i) a chronological evaluation protocol with forward-chaining cross-validation; (ii) a comparison of four regression and four classification models against a persistence baseline; (iii) a robustness analysis on the COVID-19 lockdown period and across individual cities. The remaining models

we implemented (ridge, polynomial, k-NN, MLP regression, Naive Bayes, perceptron) are reported in the appendix.

Section 2 reviews related work, Section 3 describes the data, Sections 4 and 5 the methods and set-up, Section 6 the results and Section 7 the discussion, followed by the conclusion.

2 Related Work

Kumar and Pande [3] compared five ML models for air-quality prediction on six years of data from 23 Indian cities, and Natarajan et al. [4] combined an optimisation algorithm with a decision tree for AQI prediction in major Indian cities. Mahato et al. [5] documented a sharp fall in pollutant levels in Delhi during the 2020 lockdown, which gives us a natural distribution shift to use as a stress test. For time-ordered data, Bergmeir and Benítez [6] discuss when standard and blocked cross-validation are valid; we use forward-chaining splits so that no model is trained on the future. All models were implemented with scikit-learn [7]; the algorithms follow [8].

3 Dataset and Preprocessing

3.1 Source and description

We used the file city_day.csv from the Kaggle dataset “Air Quality Data in India (2015–2020)” [9], which compiles data from the CPCB Central Control Room portal. The file has 29,531 rows and 16 columns for 26 cities, covering 1 January 2015 to 1 July 2020. The columns are city, date, twelve pollutant concentrations (PM2.5, PM10, NO, NO_2, NOx, NH_3, CO, SO_2, O_3, benzene, toluene, xylene), the AQI and the AQI category. Table 1 shows the share of missing values per variable.

Table 1. Variables and missing values in city_day.csv.

Variable	Unit	Missing (%)	Use
City, Date	categorical; day	0.0	one-hot feature; split and season
PM2.5	µg/m³	15.6	feature (day t)
PM10	µg/m³	37.7	feature (day t)
NO	µg/m³	12.1	feature (day t)
NO_2	µg/m³	12.1	feature (day t)
NOx	µg/m³	14.2	feature (day t)
NH_3	µg/m³	35.0	feature (day t)
CO	mg/m³	7.0	feature (day t)
SO_2	µg/m³	13.1	feature (day t)
O_3	µg/m³	13.6	feature (day t)

Variable	Unit	Missing (%)	Use
Benzene	µg/m³	19.0	feature (day t)
Toluene	µg/m³	27.2	feature (day t)
Xylene	µg/m³	61.3	dropped (more than 50% missing)
AQI	index 0–500	15.9	feature (t, t−1, 3-day mean); target (t+1)
AQI_Bucket	6 categories	15.9	not used (derived from AQI)

3.2 Missing values

Missingness is substantial (Fig. 1). Xylene is missing in 61.3% of rows and was dropped. PM10 (37.7%) and NH_3 (35.0%) were kept. The remaining gaps were filled with the training-set median inside the model pipeline, so test data never influence the imputation. Rows without a valid AQI on day t or day t+1 were removed.

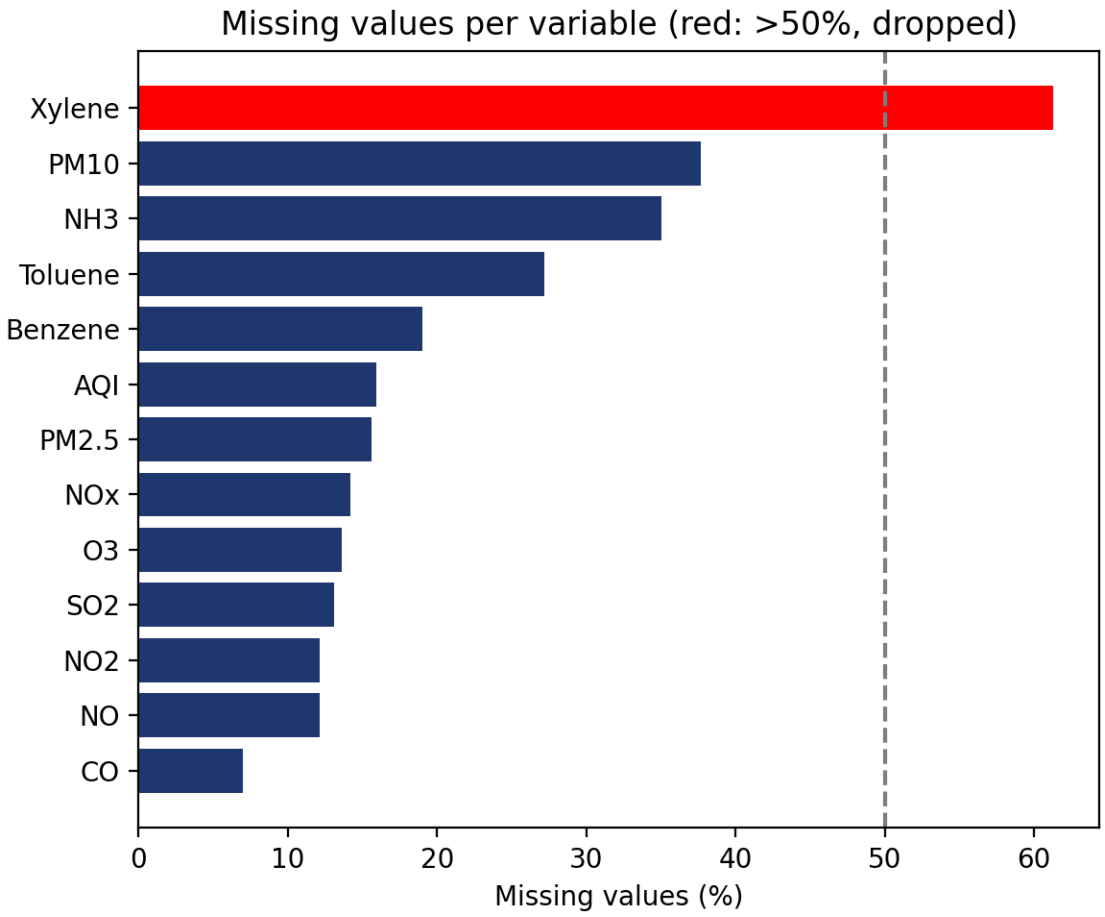


Fig. 1. Percentage of missing values per variable in city_day.csv (red: above the 50% threshold; dropped). The dashed line marks the 50% cut-off.

3.3 Feature construction, outliers and scaling

For each city the data were sorted by date and the target was defined as the AQI of the next calendar day; pairs separated by a gap were discarded. The features for day t are the 11 remaining pollutant concentrations, the AQI of day t, the AQI of day t−1, the three-day mean AQI, the month encoded as sine and cosine (to capture seasonality; Fig. 2 shows a winter rise for several cities) and the city (one-hot encoded, one indicator for each of the 18 cities seen in training).

The AQI scale ends at 500, but 543 records exceeded it; these were capped at 500. High-pollution episodes are real events rather than errors, so instead of deleting outliers we winsorised each feature at its 1st and 99th training percentiles. Features were then standardised, $z = (x - \mu)/\sigma$, with μ and σ estimated on the training data only.

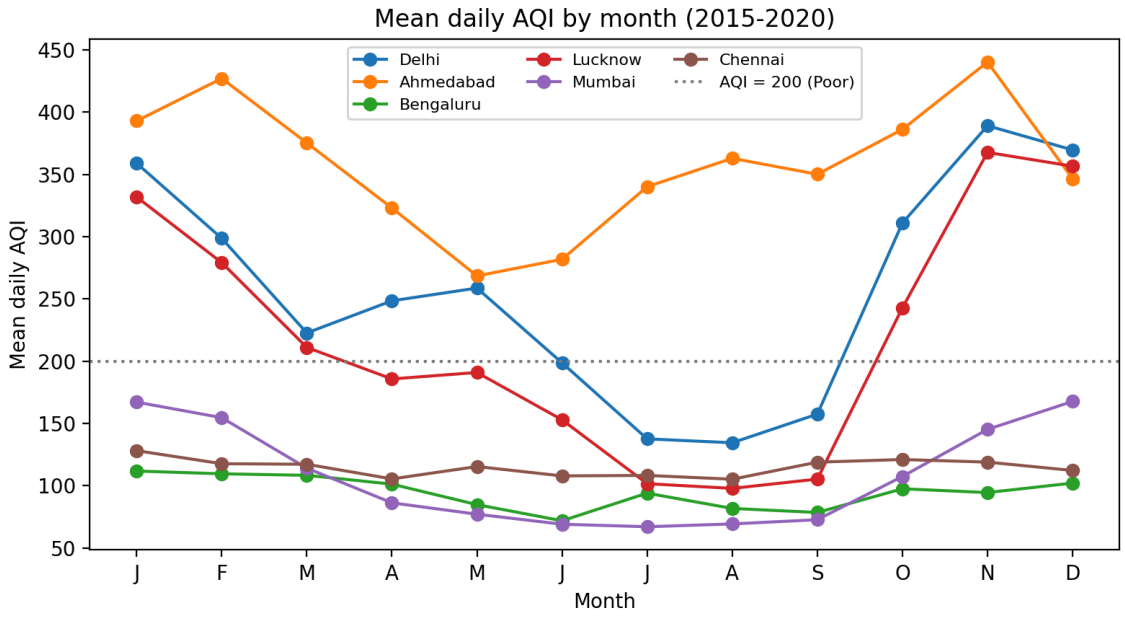


Fig. 2. Mean daily AQI by month for six cities (2015–2020), with AQI capped at 500. The dotted line marks AQI = 200, the threshold of the "Poor" category used as the health-alert label.

3.4 Feature selection and PCA

In the training years, next-day AQI was most strongly correlated with today's AQI (r = 0.92), PM10 (0.88) and PM2.5 (0.76), followed by NOx (0.52), NO (0.51) and NO_2 (0.47) (Fig. 3). Many pollutants are correlated with each other, which motivates regularised models such as lasso. PCA on the 16 numeric features needed 9 components for 90% and 12 for 95% of the variance; the first component explained 39.0% (Fig. 4).

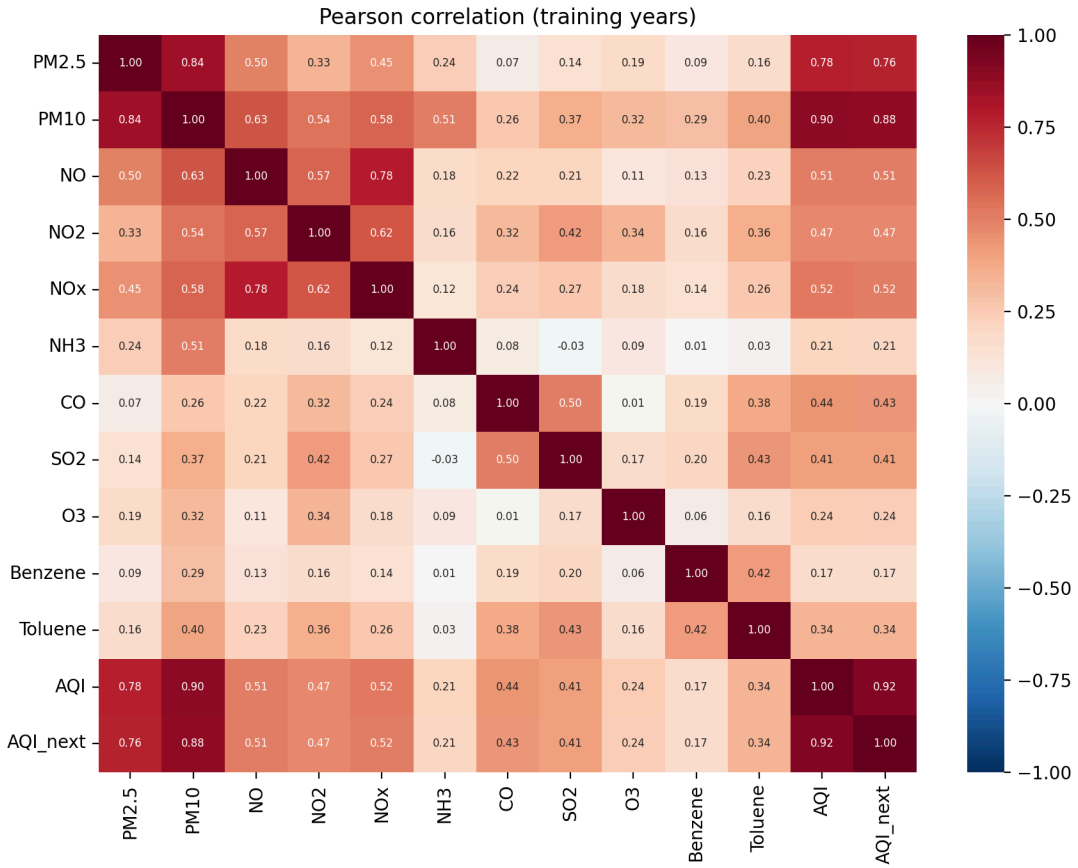


Fig. 3. Pearson correlation matrix of pollutants, current AQI and next-day AQI (training years 2015–2018).

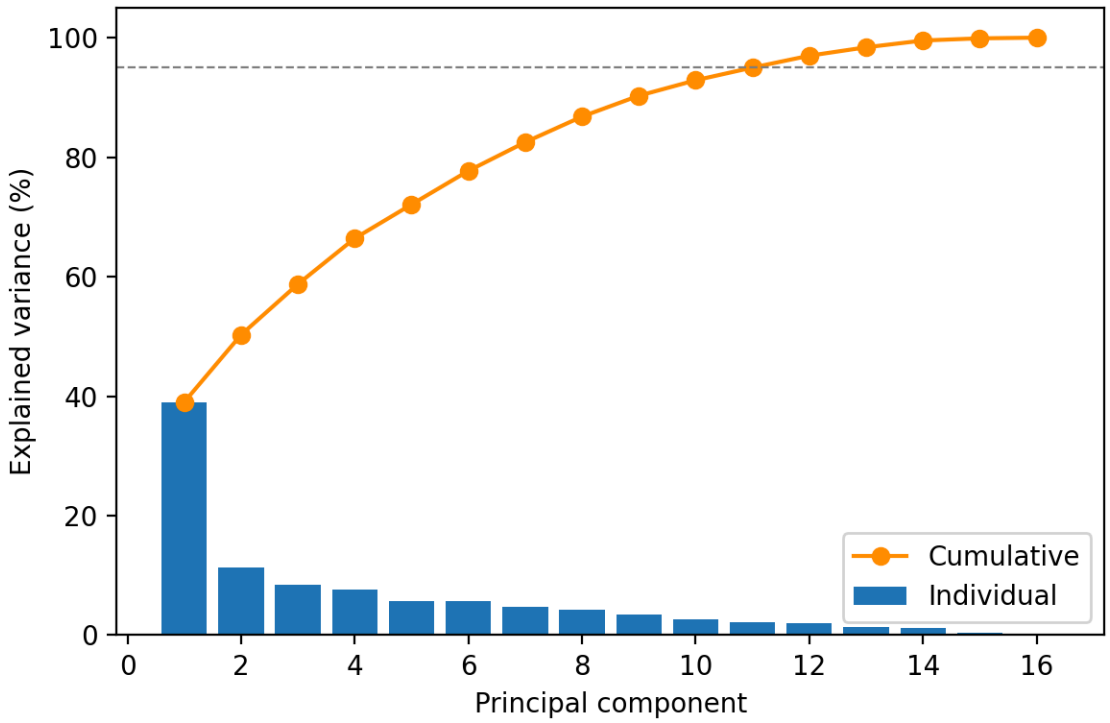
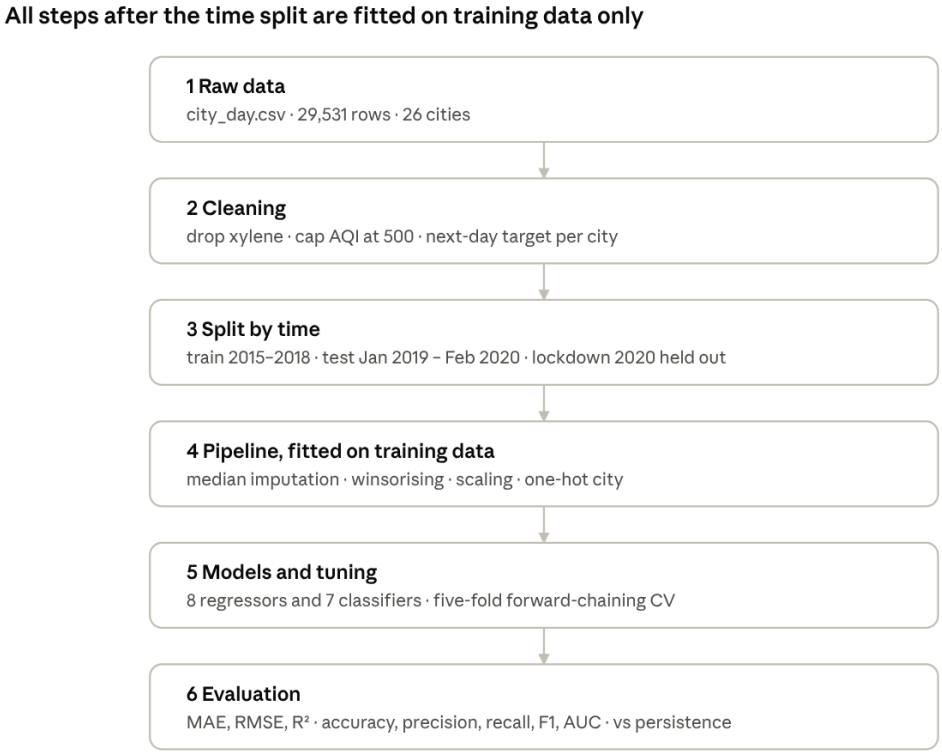


Fig. 4. Individual and cumulative explained variance of the principal components of the 16 numeric features (training data). The dashed line marks 95% cumulative variance.

4 Methodology

Fig. 5 summarises the end-to-end workflow. All steps after the chronological split are wrapped in a single scikit-learn Pipeline (median imputation, winsorising, standardisation, one-hot encoding, model) that is fitted on training data only. Let x be the feature vector of a city-day and y the next-day AQI.



methodology pipeline · 6 steps

Fig. 5. Methodology pipeline used in this study.

Persistence baseline. The naive forecast says tomorrow equals today: $\hat{y}(t+1) = AQI(t)$. For the alert task it predicts an alert when today's AQI exceeds 200. Every model must beat this baseline to be worth using.

$$\hat{y}(t+1) = AQI(t) \qquad (1)$$

4.1 Regression models

- **Linear regression (OLS)** fits $\hat{y} = w^{T}x + b$ by minimising the mean squared error (Eq. 2).
- **Lasso** adds an L1 penalty $\lambda\Sigma|w_i|$ to the squared error (Eq. 3), which shrinks weak coefficients to exactly zero and therefore acts as feature selection.
- **Decision tree regression** recursively splits the feature space to minimise within-node squared error.
- **Support-vector regression (SVR)** with an RBF kernel fits a function that ignores errors smaller than ε and penalises larger errors linearly, with regularisation parameter C (Eq. 4).

$$J(w) = (1/n)\,\Sigma_i\,(y_i - \hat{y}_i)^2 \qquad (2)$$

$$J(\text{ridge}) = J + \lambda\,\Sigma\square\, w_i^2, \quad J(\text{lasso}) = J + \lambda\,\Sigma\square\,|w_i| \qquad (3)$$

$$\min\ \tfrac{1}{2}\|w\|^2 + C\,\Sigma_i\,(\xi_i + \xi'_i), \text{ subject to } |y_i - f(x_i)| \le \varepsilon + \xi_i, \text{ with RBF kernel } K(x, x') = \exp(-\gamma\|x - x'\|^2) \qquad (4)$$

4.2 Classification models

The health-alert label is 1 if the next-day AQI is above 200 (Poor, Very Poor or Severe) and 0 otherwise.

- **Logistic regression** models $P(\text{alert} \mid x) = \sigma(w^T x + b)$, trained with L2-regularised cross-entropy (inverse strength C) (Eq. 5).
- **Decision tree** chooses splits that minimise Gini impurity.
- **Support-vector machine (SVM)** with an RBF kernel finds a soft-margin separating boundary.
- **Multilayer perceptron (MLP)** with two hidden layers (64 and 32 ReLU units), trained with the Adam optimiser, an L2 penalty and early stopping.

$P(\text{alert} \mid x) = \sigma(w^T x + b)$, $\sigma(z) = 1 / (1 + e^{(-z)})$ (5)

4.3 Evaluation metrics

Regression: $MAE = \text{mean}(|y - \hat{y}|)$, $RMSE = \sqrt{\text{mean}((y - \hat{y})^2)}$ and $R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$. Classification: accuracy, precision = TP/(TP+FP), recall = TP/(TP+FN), F1 = 2PR/(P+R) and the area under the ROC curve (AUC). A missed alert (false negative) is costlier for public health than a false alarm, so recall and F1 are emphasised over accuracy.

5 Experimental Setup

Split. To mimic real forecasting, the data were split by time and not at random. Training: 2015–2018 (13,041 samples). Test: 1 January 2019 to 29 February 2020 (8,374 samples). The national lockdown period, 25 March to 30 June 2020 (2,369 samples), was held out as a separate stress test. The alert rate was 33.1% in training, 23.2% in the test set and 1.4% during the lockdown.

Cross-validation and tuning. Hyper-parameters were chosen by grid search with five-fold forward-chaining cross-validation (TimeSeriesSplit) on the training years, optimising RMSE for regression and F1 for classification. The training data were sorted by date before splitting, and all preprocessing was refitted inside every fold to prevent leakage. Table 2 lists the grids and the selected values.

Table 2. Hyper-parameter grids and values selected by five-fold forward-chaining CV.

Model	Grid searched	Selected
Lasso	alpha {0.01, 0.1, 1, 5}	alpha = 1
Decision tree (reg.)	max depth {4, 6, 8}; min leaf {20, 50}	depth = 6, min leaf = 50
SVR (RBF)	C {10, 100}; ε {5, 15}	C = 100, ε = 15
Logistic regression	C {0.01, 0.1, 1, 10}; class weight {none, balanced}	C = 1, none
Decision tree (clf.)	max depth {4, 8, 12}; class weight {none, balanced}	depth = 4, balanced
SVM (RBF)	C {0.1, 1, 10}; class weight {none, balanced}	C = 1, none
MLP (clf.)	alpha {1e-4, 1e-2, 1}	alpha = 1

Unseen cities. Seven cities (Guwahati, Coimbatore, Chandigarh, Bhopal, Shillong, Kochi, Ernakulam) have data only after 2018, so they appear only in the test set. Their one-hot indicator is all zeros, and the model has to generalise from other cities.

Environment. Python 3.10.11, scikit-learn 1.7.2, pandas, NumPy and Matplotlib on [your laptop/OS]; random seed 42 for all stochastic models. The code in the GitHub repository regenerates every number and figure in this report.

6 Results

6.1 Next-day AQI regression

Table 3. Next-day AQI regression on the 2019–20 test set (AQI units). CV-RMSE is the mean over the five training folds. The remaining four models are in Appendix A.

Model	MAE	RMSE	R^2	CV-RMSE
Persistence baseline	25.59	42.43	0.839	–
Linear regression (OLS)	24.55	36.90	0.879	43.04
Lasso	24.39	37.75	0.873	42.26
Decision tree	23.33	36.39	0.882	44.15
SVR (RBF)	23.69	**34.96**	**0.891**	44.41

Table 3 shows that every learned model beats persistence (RMSE 42.43). The SVR achieved the lowest test RMSE (34.96, a 17.6% reduction) and the highest R^2 (0.891). The decision tree had the lowest MAE (23.33) but a higher RMSE, which suggests smaller errors on typical days but larger errors on extreme days. Plain OLS (36.90) was within 5.5% of the best model, which suggests that most of the predictable signal is linear in today's AQI and particulate levels. The ranking by cross-validation differs from the ranking on the test set: the SVR has the highest CV-RMSE of the four, and across all eight models the polynomial ridge model had the lowest CV-RMSE (41.75; Appendix A). The differences between the better models are therefore small, and "best model" should be read with some caution. Test errors were lower than CV errors, probably because the test period was less polluted (alert rate 23.2% against 33.1% in training) and large AQI values produce large absolute errors.

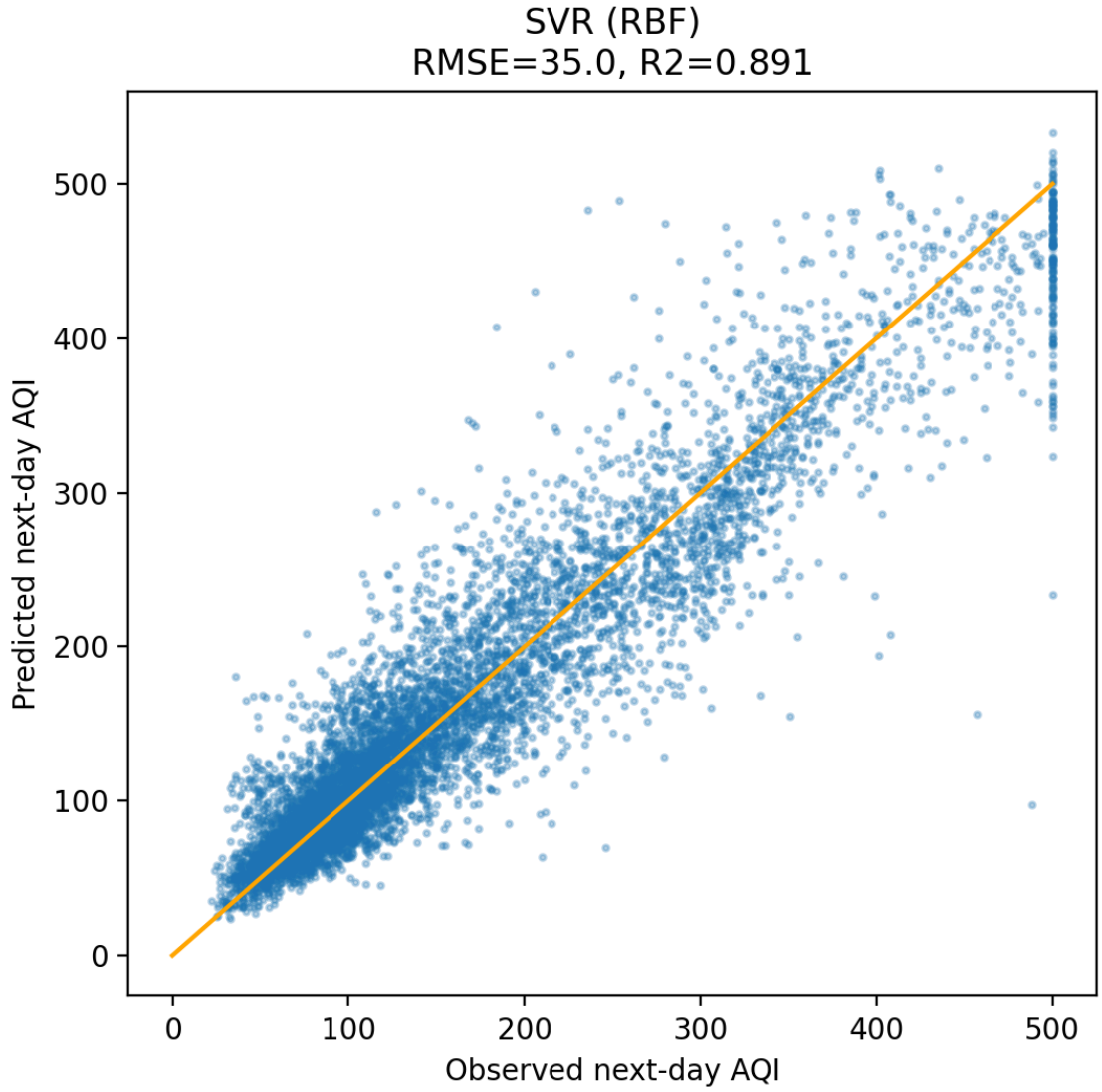


Fig. 6. Observed versus predicted next-day AQI for the SVR (RBF) on the test set; the orange line is perfect prediction. The vertical column at 500 is days capped at the AQI maximum; a few predictions exceed 500 because the model output is not bounded.

The lasso coefficients (standardised scale) ranked today's AQI (46.7), PM2.5 (30.1), the three-day mean AQI (25.9) and CO (17.0) as the strongest predictors.

6.2 Health-alert classification

Table 4. Health-alert (next-day AQI > 200) classification on the 2019–20 test set. The other three models are in Appendix B.

Model	Accuracy	Precision	Recall	F1	AUC
Persistence baseline	0.922	0.832	0.834	0.833	–

Model	Accuracy	Precision	Recall	F1	AUC
Logistic regression	0.938	0.873	0.860	**0.866**	**0.981**
Decision tree	0.927	0.791	0.931	0.855	0.974
SVM (RBF)	0.938	0.870	0.863	**0.866**	0.979
MLP (64-32)	0.937	0.865	0.865	0.865	0.981

Logistic regression, the RBF-SVM and the MLP were practically tied (F1 0.865–0.866, AUC 0.979–0.981), and each improved on persistence (F1 0.833) through higher precision and recall. The depth-4 decision tree with balanced class weights had the highest recall (0.931) at the cost of lower precision (0.791), that is, more false alarms, which may be preferable for a cautious public-health agency. For logistic regression the confusion matrix was TN = 6,187, FP = 243, FN = 273, TP = 1,671, so 273 of 1,944 real alert days (14.0%) were missed.

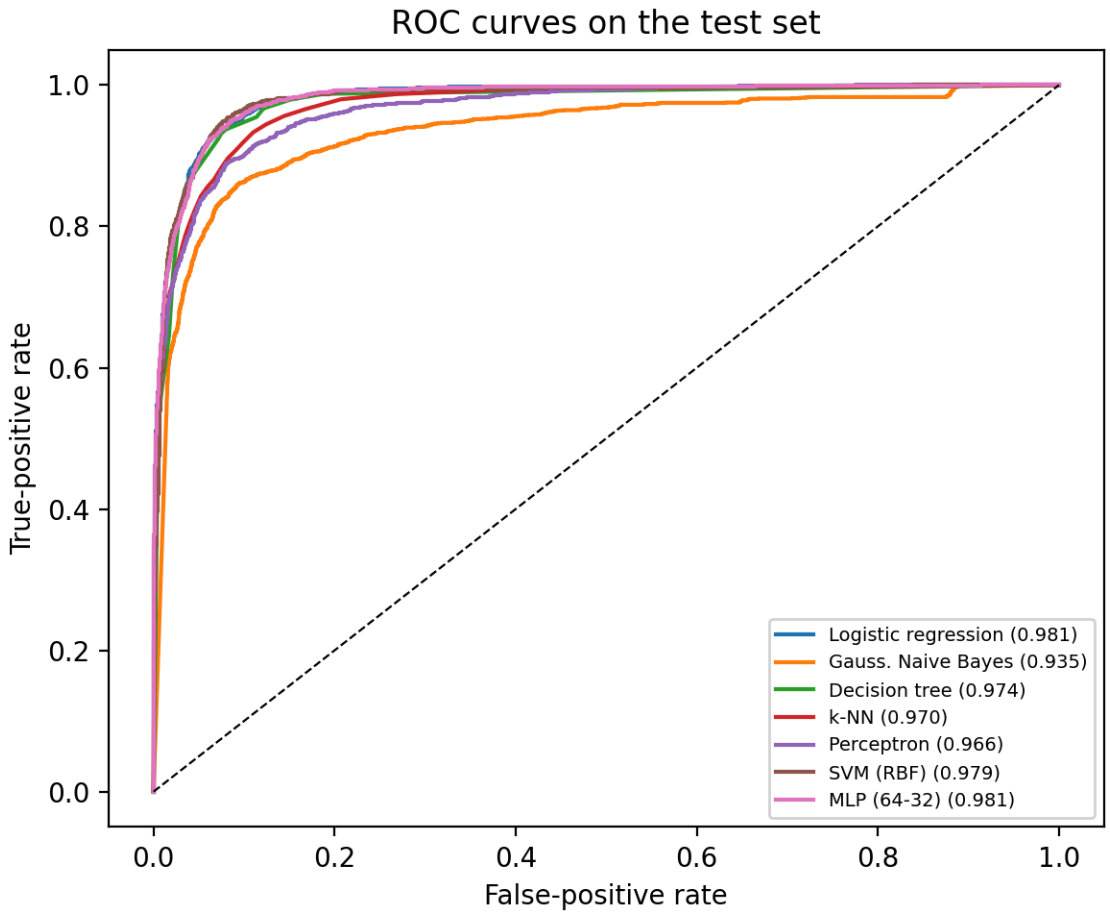


Fig. 7. ROC curves of the seven classifiers on the test set (AUC in brackets). The logistic regression, SVM and MLP curves almost overlap; the dashed diagonal is random guessing.

6.3 Effect of PCA and robustness (COVID-19 lockdown, individual cities)

Lockdown. In the lockdown period (alert rate 1.4%) the tuned SVR was clearly worse than persistence (Table 5). Only the SVR was evaluated in this period. A likely reason is that pollution fell abruptly (reported for Delhi in [5]), so relationships learned in 2015–2018 no longer held, whereas persistence uses only the latest value. This explanation was not tested.

Table 5. Next-day AQI regression during the 2020 lockdown.

Setting	MAE	RMSE	R^2
Persistence	11.53	21.11	0.734
SVR (RBF)	16.91	25.18	0.621

Individual cities. The per-city results (Appendix C) show differences between cities that the overall score hides. The SVR beat persistence in 20 of the 25 cities that have test-period samples (18 of which also appear in the training years) but was worse in five: Kolkata, Coimbatore, Shillong, Kochi and Ernakulam. Four of these five (all except Kolkata, which had 205 training samples) had no training data at all. The SVR beat persistence in the other three unseen cities (Guwahati, Chandigarh, Bhopal).

PCA. We also tried PCA, keeping 95% of the variance (Fig. 4 and Table 6). PCA did not help: the RMSE of OLS rose from 36.90 to 40.07 and that of k-NN from 40.78 to 41.70, so it was not used in the final models.

Table 6. PCA (95% of variance) versus all features, 2019–20 test set.

Setting	MAE	RMSE	R^2
OLS, all features	24.55	36.90	0.879
OLS, PCA 95%	26.42	40.07	0.857
k-NN (k=15), all features	28.46	40.78	0.852
k-NN (k=15), PCA 95%	29.09	41.70	0.845

7 Discussion

What worked. A small set of features (today's AQI, particulate levels, season and city) carries most of the information. Regularised linear and kernel models capture it with little tuning, and logistic regression reached the joint-best F1 (0.866) and has coefficients that can be inspected.

Limitations. (i) The data stop in July 2020 and contain no meteorological variables (wind, temperature, humidity, boundary-layer height), which drive day-to-day changes. (ii) City-level averages hide differences between monitoring stations. (iii) Heavy missingness was handled by simple median imputation. (iv) The lockdown result shows that models must be monitored and re-trained when emission patterns change; a fixed model should never replace expert judgement. (v) The final model was chosen on test RMSE, while cross-validation favoured a different model, so a fully nested selection would be cleaner.

Ethics and SDG impact. The data are aggregate environmental measurements with no personal information. The main ethical risk is over-reliance: a missed alert could expose vulnerable people, so a deployed system should favour recall, publish its uncertainty and be audited across cities, since smaller cities with fewer monitors may receive less accurate forecasts. A fallback to persistence may be considered for cities with no training data (the SVR was worse than persistence in 4 of the 7 such cities). Next-day alerts of this kind are relevant to SDG target 3.9 (reducing deaths and illnesses from air pollution) and SDG target 11.6 (reducing the adverse environmental impact of cities, including air quality).

8 Conclusion and Future Work

Using CPCB-derived data for 26 Indian cities and a chronological protocol, we showed that the best of the four main models per task forecast next-day AQI with R^2 up to 0.891 (SVR) and issue "Poor"-day alerts with F1 up to 0.866 (logistic regression and SVM), both better than a persistence baseline on the 2019–2020 test set. The tuned SVR was worse than persistence during the lockdown and in 5 of 25 cities, four of which had no training data. Future work will add weather forecasts, extend the data beyond 2020, use station-level data and test online re-training.

Declarations

Data availability. The dataset is publicly available at https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india (accessed [date]); the original source is the CPCB Central Control Room.

Code, figures and result tables: [https://github.com/ananya-jais/aqi-forecast].

Conflict of interest. The author declares no conflict of interest.

Funding. None.

Use of AI tools. An AI assistant (Claude, Anthropic) was used to help set up the code structure and draft parts of the report text. I ran all experiments myself, checked every number in this report against my own output files, and take full responsibility for the content.

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Appendix A. All regression models

Table A1. Next-day AQI regression, all eight models (2019–20 test set).

Model	MAE	RMSE	R^2	CV-RMSE
Persistence baseline	25.59	42.43	0.839	–
Linear regression (OLS)	24.55	36.90	0.879	43.04
Ridge	24.45	36.98	0.878	42.19
Lasso	24.39	37.75	0.873	42.26
Polynomial (deg 2) + Ridge	24.37	35.26	0.889	41.75
Decision tree	23.33	36.39	0.882	44.15
k-NN regressor	28.15	40.39	0.855	46.29
SVR (RBF)	23.69	34.96	0.891	44.41
MLP (64-32)	26.08	38.46	0.868	43.51

Appendix B. All classification models

Table B1. Health-alert classification, all seven models (2019–20 test set).

Model	Accuracy	Precision	Recall	F1	AUC
Persistence baseline	0.922	0.832	0.834	0.833	–

Model	Accuracy	Precision	Recall	F1	AUC
Logistic regression	0.938	0.873	0.860	0.866	0.981
Gaussian Naive Bayes	0.865	0.656	0.882	0.752	0.935
Decision tree	0.927	0.791	0.931	0.855	0.974
k-NN	0.924	0.857	0.807	0.831	0.970
Perceptron	0.921	0.855	0.794	0.823	0.966
SVM (RBF)	0.938	0.870	0.863	0.866	0.979
MLP (64-32)	0.937	0.865	0.865	0.865	0.981

Appendix C. Per-city test error

Table C1. Next-day AQI RMSE by city on the 2019–20 test set for the SVR and for persistence. n_train = 0 means the city was unseen in training. Cities where the SVR was worse than persistence: Kolkata, Coimbatore, Shillong, Kochi, Ernakulam.

City	n_train	n_test	Mean AQI	RMSE SVR	RMSE persistence
Bengaluru	1354	425	92	12.7	13.2
Chennai	1325	425	100	28.5	33.2
Jaipur	538	425	120	27.6	36.1
Mumbai	222	425	116	21.3	23.0
Lucknow	1332	425	208	33.6	49.1
Kolkata	205	425	152	40.3	35.8
Delhi	1446	425	238	38.1	51.3
Hyderabad	1320	425	95	13.2	14.6
Gurugram	856	423	198	41.4	53.7
Amritsar	566	416	110	30.6	39.0
Patna	872	415	220	39.9	43.8
Thiruvananthapuram	511	407	77	16.4	18.5
Ahmedabad	780	402	403	70.6	90.5
Visakhapatnam	625	382	122	23.8	32.3
Guwahati	0	368	151	51.3	66.8

City	n_train	n_test	Mean AQI	RMSE SVR	RMSE persistence
Amaravati	313	363	95	20.5	23.8
Brajrajnagar	264	352	149	32.7	41.2
Jorapokhar	283	351	159	38.8	41.8
Talcher	229	341	181	38.9	47.1
Coimbatore	0	235	78	20.7	9.5
Chandigarh	0	174	122	32.1	39.9
Bhopal	0	165	157	31.7	42.5
Shillong	0	106	61	33.3	16.1
Kochi	0	38	130	46.3	34.5
Ernakulam	0	36	102	65.8	15.4